BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Tracing Programming Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. When is a condition-controlled loop suitable?

2. Linear-search [7, 2, 9, 4] for 4. How many comparisons are needed?
   A) 1
   B) 2
   C) 4
   D) 3

3. "An IF statement must execute every possible branch regardless of its condition." Correct this. Explain why.

4. Do one bubble-sort pass from left to right, smallest to largest, on [6, 1, 5, 3]. Show the array after each comparison. Which value ends up on the right?

Today's objectives

Trace programs using sequence, selection and iteration. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Trace table
Records the current variable values after relevant instructions.

Selection tracing
Only the true branch of an IF/ELSE is followed.

Iteration tracing
Record the changes for every repeated execution.

Debugging
Compare expected and traced results to find the first incorrect step.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Trace table
Records the current variable values after relevant instructions. For example, consider a score calculator.

Selection tracing
Only the true branch of an IF/ELSE is followed. For example, consider a discount total.

Iteration tracing
Record the changes for every repeated execution. For example, consider a lesson attendance.

Debugging
Compare expected and traced results to find the first incorrect step. For example, consider a countdown.

I Do - Worked model

A score calculator is being developed. Explain how trace table and selection tracing are used. [4 marks]

- Identify where this particular program needs control or values.
- Trace table [1]; Records the current variable values after relevant instructions. [1].
- Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1].
- Give one accurate point and a linked reason for each technique.

We Do - Guided practice

1. A discount total is being programmed. Explain why trace table matters. [2 marks]

2. For a lesson attendance, explain how selection tracing is used. [2 marks]

3. A countdown needs iteration tracing and debugging. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the value printed after the loop.

```
totalGB = 0
for i = 1 to 3
totalGB = totalGB + i
next i
print(totalGB)
```

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how selection tracing and iteration tracing can help a points accumulator. [4 marks]

Q4. Explain the roles of trace table, iteration tracing and debugging when creating a fitness tracker. [6 marks]

Exit Ticket - True or False?

A trace table records the values of variables after relevant instructions execute.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. When repetition continues or ends according to a condition, such as valid input.

2. C - 4

3. It chooses a branch using a condition. Selection can be nested.

4. (a) [[1, 6, 5, 3], [1, 5, 6, 3], [1, 5, 3, 6]]; largest value 6 is at the right end.

I Do - Explain the model

Identify where this particular program needs control or values.

Trace table [1]; Records the current variable values after relevant instructions. [1].

Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1].

Give one accurate point and a linked reason for each technique.

We Do - Model answers

1. Trace table [1]; Records the current variable values after relevant instructions. [1].

2. Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1].

3. Iteration tracing [1]; Record the changes for every repeated execution. [1]. Debugging [1]; Compare expected and traced results to find the first incorrect step. [1].

Code First - Answer

```
totalGB = 0
for i = 1 to 3
totalGB = totalGB + i
next i
print(totalGB)
```

Expected output: 6

You Do - Self-mark

Q2: Trace table [1]; Iteration tracing [1].

Q3: Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1]. Iteration tracing [1]; Record the changes for every repeated execution. [1].

Q4: Trace table [1]; Records the current variable values after relevant instructions. [1]. Iteration tracing [1]; Record the changes for every repeated execution. [1]. Debugging [1]; Compare expected and traced results to find the first incorrect step. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.